Retail Sales Analytics

Project & Dashboard Report

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About the project

I analysed 82,950 line items of retail sales from 27,666 transactions, 2,000 customers and 41 products (January 2022 to December 2023) to answer three questions: what drives sales, how promotions perform, and how sales move over time.

Tools: Python, Pandas, Matplotlib, SQL (SQLite), Excel. A small Flask chatbot answers common questions on the same data.

Code and data:
github.com/Ishdeeps20/Retail-Sales-Analytics-and-Interactive-Chatbot

Headline findings

1. Total sales of **Rs 124.7 million**; average basket **Rs 4,507**.

2. The top 10 products bring **44%** of sales. Units sold per product are almost equal, so revenue follows **price, not volume**.

3. Promo lines are **20%** of line items but **16.8%** of sales, with no gain in units per line (5.48 vs 5.50).

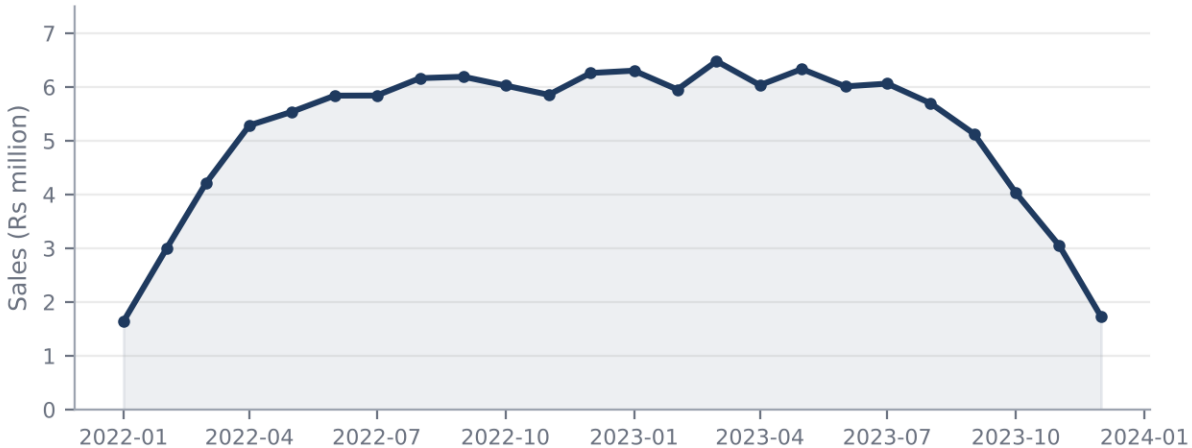
4. Monthly sales plateau at Rs 5.3–6.5M (Apr 2022 – Jul 2023); the ramps at both ends need checking (see page 4).

Retail Sales Dashboard

Jan 2022 – Dec 2023 | 82,950 line items

Rs 124.7M	27,666	2,000	Rs 4,507	41	456,026
Total sales	Transactions	Customers	Avg basket	Products	Units sold

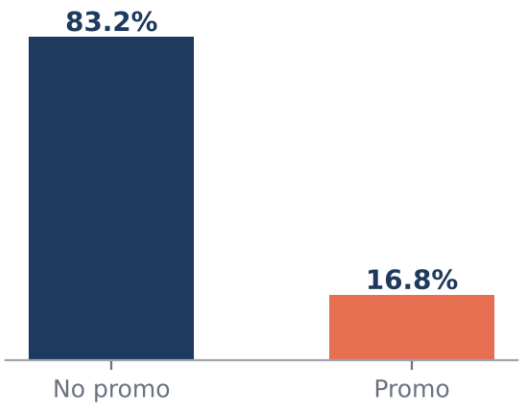
Monthly sales: ramp-up, plateau, ramp-down



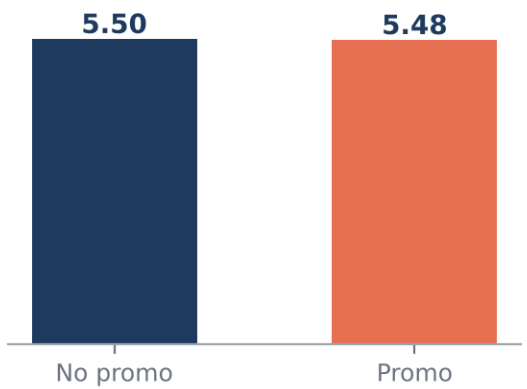
Top 10 products by sales (Rs million)



Share of sales



Avg units per line



Sales follow price, not volume (41 products)



Source: Sales_data.xlsx (github.com/Ishdeeps20/Retail-Sales-Analytics-and-Interactive-Chatbot). Built with Python, Pandas and Matplotlib.

Findings and recommendations

Revenue is driven by price, not volume	**Evidence.** Units sold are almost identical across all 41 products (about 10,700 to 11,600 each), but sales range from Rs 0.4M to Rs 6.3M. Average price and total sales are almost perfectly related (correlation 1.00); units and sales are not (-0.11).	**Recommendation.** Prioritise stock, shelf space and marketing for higher-priced, high-revenue lines such as Godrej Shampoo, Milky Mist Paneer and Santoor Soap.
Promotions are not visibly lifting volume	**Evidence.** Promo lines have a lower average unit price (Rs 228 vs Rs 285) but the same average units per line (5.48 vs 5.50).	**Recommendation.** Before expanding discounts, run a controlled comparison of the same products with and without a promo.
Sales are spread across a broad customer base	**Evidence.** The top 10% of customers account for only 17.9% of sales, and average basket is Rs 4,507.	**Recommendation.** Target retention and basket-size offers at the whole base; there is no heavy dependence on a few customers.
Demand is steady through the middle of the period	**Evidence.** From Apr 2022 to Jul 2023 monthly sales stay between Rs 5.3M and Rs 6.5M, peaking in Mar 2023.	**Recommendation.** Use this plateau as the baseline for targets and forecasts.

Method, data quality and limitations

Method

1. Loaded the Excel file with Pandas and checked structure and types.

2. Checked for missing values and duplicate rows.

3. Converted transaction month to dates and aggregated by month, product, promo flag and customer.

4. Loaded the data into SQLite and used SQL (GROUP BY, ORDER BY) to confirm the top products.

5. Built the dashboard charts in Matplotlib and wrote the findings.

Data quality checks

No missing values and no duplicate rows in 82,950 lines.

Sales amount equals unit price multiplied by units on every row.

Each line is a product in a transaction, so the data has 3 levels: customer, transaction and product.

Data dictionary is on Sheet2 of the Excel file.

Limitations

Ramp-up and ramp-down. Sales climb from Rs 1.6M (Jan 2022) and fall to Rs 1.7M (Dec 2023). This may be a coverage effect of the dataset, not a real change in demand, so it should be confirmed with the data owner.

Promotions. The comparison shows association, not cause: promo and non-promo lines differ in product mix and price.

Single source. No cost, margin or customer details, so profit and segments were not analysed.